Saroj Khadka

Graduate Student Researcher

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Summary

Biomedical Science PhD candidate with strong research experience in bacterial virulence mechanisms, host-pathogen interactions, and microbial gene regulation. Skilled in molecular biology and microbiological techniques, mentoring and science communication. Motivated to build an independent career in biomedical research.

Education

PhD **University of Pittsburgh**, PhD in Microbiology and Immunology – Pittsburgh, PA, USA 2024 – 2026

- Thesis: Mechanisms and Regulation of Klebsiella pneumoniae CPS Chain Length and Attachment.
- Advisor: Dr. Laura A. Mike

MSc **Tribhuvan University**, MSc in Medical Microbiology – Kathmandu, Nepal 2017 – 2019

- Thesis: Susceptibility to Fluoroquinolones and gyrA ser83 Mutation in Salmonella enterica serovar Typhi at a Referral Hospital of Kathmandu Valley.
- Advisor: Dr. Megharaj Banjara

Experience

Division of Infectious Diseases, Department of Medicine, Graduate Student Researcher – University of Pittsburgh, PA 2024 – 2026

- Investigate molecular mechanisms regulating Klebsiella pneumoniae capsular polysaccharide attachment and chain length.
- Develop and optimize experimental techniques for bacterial exopolysaccharide research.
- Present research findings at departmental, regional, and national scientific conferences.
- Mentor undergraduate researchers in laboratory techniques, experimental workflows, and project design.

B-Sure Path Lab & Diagnostic Centre, Molecular Biologist – Biratnagar, Nepal 2020 – 2021

- Performed routine qRT-PCR testing for COVID-19 diagnosis from clinical specimens.
- Trained newly recruited laboratory technicians in molecular diagnostic techniques and workflows.
- Managed patient records, diagnostic reporting, and laboratory quality control in compliance with National Public Health Laboratory (Nepal) guidelines.
- Coordinated diagnostic reporting with the Ministry of Health COVID-19 Crisis Management Center.

Volunteer

People's Climate March, Lead Organizer – Zurich, Switzerland Apr 2014 – July 2015

Lead organizer for the New York City branch of the People's Climate March, the largest climate march in history.

- Awarded 'Climate Hero' award by Greenpeace for my efforts organizing the march.
- Men of the year 2014 by Time magazine

Awards

AHA Predoctoral Fellow Jan 2022

The Nobel Prizes are five separate prizes that, according to Alfred Nobel's will of 1895, are awarded to 'those who, during the preceding year, have conferred the greatest benefit to humankind.'

American Heart Association

Graduate Student Researcher 2029

Awarded for outstanding scientific achievement

German Physical Society

Publications

Zur Elektrodynamik bewegter Körper
It concerned an interpretation of the Michelson–Morley experiment and the properties of light and time. Special relativity incorporates the principle that the speed of light is the same for all inertial observers regardless of the state of motion of the source.
Albert Einstein
en.wikisource.org/wiki/Translation:On_the_Electrodynamics_of_Moving_Bodies

Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt
In the second paper, he applied the quantum theory to light to explain the photoelectric effect. In particular, he used the idea of light quanta (photons) to explain experimental results, but stressed the importance of the experimental results. The importance of his work on the photoelectric effect earned him the Nobel Prize in Physics in 1921.
Albert Einstein
de.wikisource.org/
wiki/%C3%9Cber_einen_die_Erzeugung_und_Verwandlung_des_Lichtes_betreffenden_heuristischen_Gesichtspunkt

Die Grundlage der allgemeinen Relativitätstheorie
The publication of the theory of general relativity made him internationally famous. He was professor of physics at the universities of Zurich (1909–1911) and Prague (1911–1912), before he returned to ETH Zurich (1912–1914).
Albert Einstein
de.wikisource.org/wiki/Die_Grundlage_der_allgemeinen_Relativit%C3%A4tstheorie

Skills

Physics

Languages

German
Native speaker

English
Fluent

Interests

Physics

Certificates

Machine Learning Jan 2018

Quantum Computing Jan 2018

Quantum Information Jan 2018

Projects

Quantum Computing Jan 2018 – Jan 2018
Quantum computing is the use of quantum-mechanical phenomena such as superposition and entanglement to perform computation. Computers that perform quantum computations are known as quantum computers.

- Quantum Teleportation
- Quantum Cryptography

References

Professor John Doe

Professor Jane Smith